BT603 Solved Final Term paper

1. **Fermentation actually comes from Latin word "fervere" which means "to boil"**
2. Fermentation can be classified into different types on the basis of followings:
    a. *Oxygen (O_2) Need*
    b. *Production Scale*
    c. State/Mixing
    d. All of above
3. **Pilot and industrial scale fermentations** are normally carried out in stirred, aerated vessels
4. **Examples of Bioweapons**
    a. Smallpox
    b. Anthrax
    c. Botulinum toxin
    d. All of above
5. **Constitutive enzyme:** An enzyme that is synthesized continually regardless of growth conditions.
6. In 1900-1940 Citric acid is produced
7. **On the basis of state or mixing fermentation can be Shake-Flask Fermentation**
8. **Downstream Processing** includes the **recovery of the products** in a pure state and the **effluent treatment**.
9. **A medium with a high viscosity will also need a higher power input for effective stirring**
10. **Reducing media contain ingredients, such as sodium thioglycolate, that chemically combine with dissolved oxygen and deplete the oxygen in the culture medium.**
11. **Cellular yield coefficient of n-alkanes 1.03**
12. ***Carbohydrate composition of barley malt (expressed as % dry weight of total) of Reducing sugars 3-4***
13. ***Trace element Mn influencing primary and secondary metabolites production of Bacitracin***
14. **In processes used for the production of glutamic acid, limited concentrations of biotin must be present in the medium**
15. **Phenylacetic acid is still the most widely used precursor in penicillin production.**
16. **potential chlorination inhibitors, e.g. bromide,**
17. **heterologous proteins may show some degree of toxicity to the host and have a major influence on the stability of heterologous protein expression.**

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18. **The individual components of the medium can influence the viscosity of the final medium and its subsequent behaviour with respect to aeration and agitation.**
19. **Trace elements were increased; insulin, transferrin, albumin and liposomes were not needed included Co**
20. **The foaming can cause removal of cells from the medium which will lead to autolysis**
21. **The foam is unavoidable and antifoam should be used.**
22. **many antifoams are of low solubility**
23. **independent variable (nutrient, antifoam, pH, temperature, etc.)**
24. **When more than five independent variables are to be investigated, the Plackett-Burman design may be used**
25. **The oxygen demand of an industrial fermentation process is normally satisfied by aerating and agitating the fermentation broth.**
26. **The solubility of oxygen in pure water is 8 mg/L at 4°C.**
27. **Production of 100 grams of biomass from hydrocarbon requires approximately three times the amount of oxygen to produce the same amount of biomass from carbohydrate.**
28. **maximum biomass production may be achieved by satisfying the organism's maximum specific oxygen demand by maintaining the dissolved oxygen concentration greater than the critical level**
29. **Sulphite Oxidation technique does not require the measurement of dissolved oxygen concentrations**
30. **Two methods have been employed to achieve this lowering of the dissolved oxygen concentration - the static method and the dynamic method.**
31. **(NCTC) abbreviation of? National Collection of Type cultures**
32. **The isolation method most commonly used to get pure cultures is the streak plate method**
33. **lyophilization (freeze-drying)**
34. **the process of strain improvement involves the continual genetic modification of the culture,**
35. **Protoplast fusion has been demonstrated as an efficient way to induce hetero-karyon formation and recombination with high frequency.**
36. **Auxotrophy is the inability of an organism to synthesize a particular organic compound required for its growth**
37. **The quantity of inoculum normally used is between 3 and 10% of the medium volume**

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1. SSF is also called as **Koji Fermentation.**
2. **Constitutive enzyme:** An enzyme that is synthesized continually regardless of growth conditions.
3. **On the basis of state or mixing fermentation can be**
   1) Static Fermentation
   2) Shake-Flask Fermentation
   3) Stirred Fermentation.
4. Finally, the ***effluents are treated*** by chemical, physical or biological methods.
5. Crude media is made up of unrefined agricultural products e.g. containing barley.
6. **Parton and Willis (1990) advocated the use of the carbon dioxide production rate (CPR) as a transfer criterion which requires analysis of the fermenter effluent air.**
7. **The quantity of inoculum normally used is between 3 and 10% of the medium volume.**
8. **Auxotrophy is the inability of an organism to synthesize a particular organic compound required for its growth.**
9. **the process of strain improvement involves the continual genetic modification of the culture, followed by reappraisals of its cultural requirements**
10. **Fluorescence in situ hybridization (FISH)**
11. ***Streak plate method for pure culturing Of required microorganism is done for nonfilamentous unicellular microorganisms only.***
12. **Selective pressure may be used in the isolation of organisms which will grow on particular substrates, in the presence of certain compounds or under cultural conditions adverse to other types.**
13. **he oxygen balance technique appears to be the simplest method for the assessment of *KL*a and has the advantage of measuring aeration efficiency during a fermentation.**
14. **Dynamic Method of Gassing-out Respiratory activity of a growing culture in the fermenter is utilized to lower the oxygen level prior to aeration.**
15. **Two methods have been employed to achieve this lowering of the dissolved oxygen concentration - the static method and the dynamic method.**
16. **Sulphite Oxidation Technique This technique does not require the measurement of dissolved oxygen concentrations**
17. **If the *KLa* is such that the oxygen demand of the organism can be easily met the dissolved oxygen concentration will be greater than Ccrit.**
18. **The volumetric mass-transfer coefficient is used as a measure of the aeration capacity of a fermenter**
19. **The rate of movement of an oxygen molecule through the**
20. **bulk liquid is dependent on the degree of mixing (relative to the volume of the reactor) and viscosity of the medium.**
21. **Pilot- and industrial-scale fermentations are normally carried out in stirred, aerated vessels (fermenters).**

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22. **From Darlington's and Johnson's equations (in above mentioned Table) it may be seen that the production of 100 grams of biomass from hydrocarbon requires approximately three times the amount of oxygen to produce the same amount of biomass from carbohydrate.**
23. **The solubility of oxygen in pure water is 8 mg/L at 4oC.**
24. **The oxygen demand of an industrial fermentation process is normally satisfied by aerating and agitating the fermentation broth.**
25. **Medium optimization by the classical method of changing one independent variable (nutrient, antifoam, pH, temperature, etc.) while fixing all the others at a certain level**
26. **can be extremely time consuming and expensive for a large number of variables.**
27. **Because many antifoams are of low solubility they need a carrier such as lard oil, liquid paraffin or castor oil, which may be metabolized and affect the fermentation process.**
28. **Should have no effect on oxygen transfer.**
29. **The most common cause of foaming is due to proteins**
30. **in the medium, such as corn-steep liquor, meal, soybean meal, yeast extract or meat extract.**
31. **It is sometimes forgotten that oxygen, although not added to an initial medium as such, is nevertheless a very important component of the medium in many processes, and its availability can be extremely important in controlling growth rate and metabolite production.**
32. **Methylotrophic yeasts such as *Hansenula polymorpha* and *Pichia pastoris.***
33. **One unusual application of an inducer is the use of yeast mannan in streptomycin production.**
34. **A number of studies have been made with potential chlorination inhibitors, e.g. bromide, to minimize chlortetracycline production during a tetracycline fermentation.**
35. **Having established that the activity of penicillin lay in the side chain, and that the limiting factor was the synthesis of the side chain, it became standard practice to add sidechain precursors to the medium, in particular phenylacetic acid.**
36. **Many media are buffered at about pH 7.0 by the incorporation of calcium carbonate.**
37. Bacitracin Mn
38. **A typical oil contains approximately 2.4 times the energy of glucose on a per weight basis.**

Questions

Q:- what is Site-Directed Mutagenesis? (2)

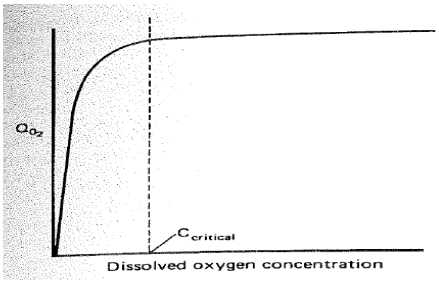
- Site-specific mutation commonly involve using mutagenic oligonucleotides in a primer extension reaction with DNA polymerase.
- This method allows for point mutation, or deletion or insertion of small stretches of DNA to be introduced at specific sites.

Q: Two examples of Methylotrophic yeasts?

Methylotrophic yeasts such as *Hansenula polymorpha* and *Pichia pastoris*

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Q: Draw a graph showing The effect of dissolved oxygen concentration on the Q_{O2} of a microorganism?



Q: Define Henry's Law?

Henry's Law describes the solubility of O_2 in nutrient solution in relation to the O_2 partial pressure in the gas phase

$$C^* = \frac{P_o}{H}$$

C* is the oxygen saturation concentration of the nutrient solution,

P_o is the partial pressure of the gas in the gas phase

H is Henry's constant, which is specific for the gas and the liquid phase.

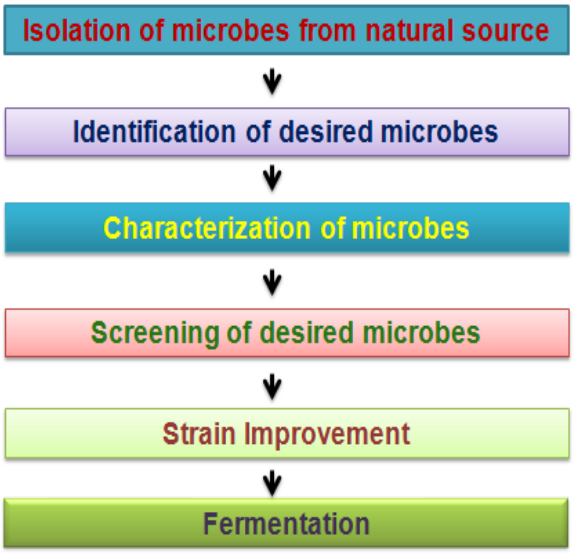
Q: Bull *et al.* (1979) cited criteria in the choice of process organism? (10)

Q: What is **Replica Plating Method? (5)**

Mutants can be detected by selecting or testing for an altered phenotype. Positive selection involves the selection of mutant cells and rejection of non-mutated cells. To identify mutants that have gained some physical trait, Replica plating is used for negative selection. Negative selection identifies mutants that have lost some physical trait - for example, auxotrophs that have nutritional requirements not possessed by the parent (nonmutated) cell.

Q: Draw a flow chart for INDUSTRIAL STRATEGY FOR USAGE OF MICROORGANISMS? (5)

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Q: **Phenotypic characteristics to characterize and identify microorganisms (3)?**

- **Microscopic morphology**
- **Metabolic differences (Biochemical Testing)**
- **Serology**
- **Fatty acid analysis**

Q: Name Trace elements use in fermentation media, any five (3)

The role of trace elements in medium formulation can be significant. Fe, Zn, Cu, Se, Mn, Mo and V. AI, Ag, Ba, Br, Cd, Co, Cr, F, Ge, J, Rb, Zr, Si, Ni and Sn

Phenotypic technique?

Genotypic technique?

Describe henry law?

What is bull criteria? 10

Name of specific inhibitors?

Name of specific inhibitors

Method of screening antibiotics?

Why antifoam is excess in media give three methods?

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Draw oxygen limitation in dynamic technique(2)

Names of 2 yeast (2)

Name Trace elements use in fermentation media, any five (3)

Write oxygen of oxygens demand during fermentation (3)

What is site directed mutagenesis (2)

Replication technique (5)

Brief note on PH of fermentation (5)

Draw flowchart of strategy of industrial microbial isolation (5)

Draw flowchart of inoculum development (10)

Names of Chemical Mutagens

Draw graph of dissolve oxygen concentration to QO2

Draw Graph of dissolve oxygen concentration to the time taken in the process.

Name the techniques used to transfer genetic material into the organism.

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